

Kianté Brantley | Curriculum Vitae

(Last updated: December 11, 2024)

Harvard University
School of Engineering and Applied Sciences
150 Western Ave
Allston, MA 02134

Pronouns: [he/him/his](#)
Website: [xkianteb.github.io](#)
Email: kdbrantley@g.harvard.edu
Github: [xkianteb](#)

Research Employment

2024 – **Assistant Professor.**
Kempner Institute and School of Engineering and Applied Sciences (SEAS), Harvard University.

2022 – 2024 **Postdoctoral Associate.**
Department of Computer Science, Cornell University

Microsoft Research, Summer Intern.

2021 Microsoft Research, Redmond
Host(s): Yizhe Zhang, Michel Galley, Bill Dolan

2020 Microsoft Research, Montréal
Host(s): Geoffrey J. Gordon and Soroush Mehri

2019 Microsoft Research, NYC
Host(s): Miro Dudík and Hal Daumé III

2018 Microsoft Research, NYC
Host(s): Miro Dudík and Hal Daumé III

Education

2016 – 2022 **University of Maryland, College Park.**
Ph.D., Computer Science (advisor: Hal Daumé III).
Thesis title: *Expert-In-The-Loop for Sequential Decisions and Predictions*.

2018 – 2022 **New York University.**
Visiting Researcher, Computer Science (host: Kyunghyun Cho).

2015 – 2016 **University of Maryland, Baltimore County.**
M.Sc. Computer Science (advisor: Tim Oates).
Thesis title: *BCAP: A Pruning Technique to Reduce Overfitting*.

2013 – 2015 **University of Maryland, Baltimore County.**
B.Sc. Computer Science.
Minor in Mathematics.

2011 – 2013 **The Community College of Baltimore County.**
A.A. Computer Science and A.A. Mathematics.
Minor in Mathematics.

Research Publications

Thesis

1. **Brantley, K.** *Expert-in-the-Loop for Sequential Decisions and Predictions* PhD thesis (University of Maryland, College Park, 2021).
2. **Brantley, K.** *BCAP: An Artificial Neural Network Pruning Technique to Reduce Overfitting* MA thesis (University of Maryland, Baltimore County, 2016).

Journal Articles

1. Wu, R., Chen, Y., Swamy, G., **Brantley, K.**, Sun, W., Diffusing States and Matching Scores: A New Framework for Imitation Learning (2024).

Conference Papers

1. **Brantley, K.**, Fang, Z., Dean, S., Joachims, T., *Ranking with Long-Term Constraints in Proceedings of the ACM International Conference on Web Search and Data Mining (WSDM)* (2024), 47–56.
2. Chang, J. D., Sreenivas, D., Huang, Y., **Brantley, K.**, Sun, W., *Adversarial Imitation Learning via Boosting in The International Conference on Learning Representations (ICLR)* (2024).
3. Gao, Z., Chang, J. D., Zhan, W., Oertell, O., Swamy, G., **Brantley, K.**, Joachims, T., Bagnell, J. A., Lee, J. D., Sun, W., *REBEL: Reinforcement Learning via Regressing Relative Rewards in Preprint* (2024).
4. Phan, M., **Brantley, K.**, Milani, S., Mehri, S., Swamy, G., Gordon, G. J., *When is Transfer Learning Possible? in Proceedings of the International Conference on Machine Learning (ICML)* (2024).
5. Tucker, A. D., **Brantley, K.**, Cahall, A., Joachims, T., *Coactive Learning for Large Language Models using Implicit User Feedback in Proceedings of the International Conference on Machine Learning (ICML)* (2024).
6. Faltings, F., Galley, M., **Brantley, K.**, Peng, B., Cai, W., Zhang, Y., Gao, J., Dolan, W. B., *Interactive Text Generation in Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)* (2023), 4450–4468.
7. Ramamurthy, R., Ammanabrolu, P., **Brantley, K.**, Hessel, J., Sifa, R., Bauckhage, C., Hajishirzi, H., Choi, Y., *Is Reinforcement Learning (Not) for Natural Language Processing: Benchmarks, Baselines, and Building Blocks for Natural Language Policy Optimization in The International Conference on Learning Representations (ICLR) Spotlight* (2023).
8. Wu, A., **Brantley, K.**, Kojima, N., Artzi, Y., *lilGym: Natural Language Visual Reasoning with Reinforcement Learning in Proceedings of the Conference of the Association for Computational Linguistics (ACL)* (2023).
9. **Brantley, K.**, Mehri, S., Gordon, G. J., *Successor feature sets: Generalizing successor representations across policies in Proceedings of the National Conference on Artificial Intelligence (AAAI)* **35** (2021), 11774–11781.
10. **Brantley, K.**, Dudik, M., Lykouris, T., Miryoosefi, S., Simchowitz, M., Slivkins, A., Sun, W., *Constrained episodic reinforcement learning in concave-convex and knapsack settings in Advances in Neural Information Processing Systems (NeurIPS)* **33** (2020), 16315–16326.
11. **Brantley, K.**, Sharaf, A., Daumé III, H., *Active Imitation Learning with Noisy Guidance in Proceedings of the Conference of the Association for Computational Linguistics (ACL)* (2020), 2093–2105.
12. **Brantley, K.**, Sun, W., Henaff, M., *Disagreement-regularized imitation learning in The International Conference on Learning Representations (ICLR) Spotlight* (2019).
13. Miryoosefi, S., **Brantley, K.**, Daumé III, H., Dudik, M., Schapire, R. E., *Reinforcement learning with convex constraints in Advances in Neural Information Processing Systems (NeurIPS)* **32** (2019).
14. Welleck, S., **Brantley, K.**, Iii, H. D., Cho, K., *Non-monotonic sequential text generation in Proceedings of the International Conference on Machine Learning (ICML)* (2019), 6716–6726.

Workshop Papers

1. Chang, J., **Brantley, K.**, Ramamurthy, R., Misra, D., Sun, W., *Learning to Generate Better Than Your LLM in NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following* (2023).
2. Gao, G., Chang, J. D., Cardie, C., **Brantley, K.**, Joachims, T., *Policy-Gradient Training of Language Models for Ranking in NeurIPS 2023 Workshop Foundation Models for Decision Making Workshop* (2023).
3. Sharaf, A., Feng, S., Nguyen, K., **Brantley, K.**, Daumé III, H., *The UMD Neural Machine Translation Systems at WMT17 Bandit Learning Task in Proceedings of the Second Conference on Machine Translation* (2017), 667–673.
4. Ganesan, A., **Brantley, K.**, Pan, S., Chen, J., *Ldaexplore: Visualizing topic models generated using latent dirichlet allocation in extvis Workshop - Intelligent User Interfaces (IUI)* (2015).

Preprint Papers

1. Chang, J. D., Shan, W., Oertell, O., **Brantley, K.**, Misra, D., Lee, J. D., Sun, W., *Dataset Reset Policy Optimization for RLHF in Preprint* (2024).
2. Gao, Z., **Brantley, K.**, Joachims, T., *Reviewer2: Optimizing Review Generation Through Prompt Generation in Preprint* (2024).
3. Gao, Z., Zhan, W., Chang, J. D., Swamy, G., **Brantley, K.**, Lee, J. D., Sun, W., *Regressing the Relative Future: Efficient Policy Optimization for Multi-turn RLHF in Preprint* (2024).
4. Monea, G., Bosselut, A., **Brantley, K.**, Artzi, Y., *LLMs Are In-Context Reinforcement Learners in Preprint* (2024).
5. Oertell, O., Chang, J. D., Zhang, Y., **Brantley, K.**, Sun, W., *RL for Consistency Models: Faster Reward Guided Text-to-Image Generation in Preprint* (2024).
6. Wu, A., **Brantley, K.**, Artzi, Y., *A Surprising Failure? Multimodal LLMs and the NLVR Challenge in Preprint* (2024).

Fellowships

2022 – 2024	Computing Research Association (CRA) Computing Innovation NSF Computing and Information Science and Engineering (CISE) Role: Postdoctoral Associate \$255k
2021	Ann G. Wylie Dissertation Fellowship Graduate School's Semester Dissertation Fellowship Role: Graduate Fellow \$15k
2020	Microsoft Dissertation Research Grant Role: Graduate Fellow \$25k
2018 – 2020	ACM SIGHPC/Intel Computational and Data Science Fellowships Role: Graduate Fellow \$30k
2017 – 2020	Sloan Research Fellowship Role: Graduate Fellow \$15k
2016 – 2018	Louis Stokes Alliance for Minority Participation Bridge to the Doctorate Program (LSAMP BD) Fellowship NSF Standard Grant Role: Graduate Fellow \$80k
2016 – 2018	University of Maryland College Park Dean's Fellowship Role: Graduate Fellow \$5k
2015	Department of Defense Graduate Fellowship Role: Graduate Fellow \$40k

Honors and Awards

2021	Spotlight talk , Empirical Methods in Natural Language Processing
2021	Spotlight talk , International Conference on Learning Representations
2021	Spotlight talk , International Conference on Learning Representations
2021	2nd place , The New York Academy of Sciences Star Talks
2013 – 2015	Transfer-Scholarships in Information Technology and Engineering (T-SITE) Scholar
2013 – 2015	Maryland Senators and Delegates Scholarship
2013 – 2015	Transfer Student Alliance Scholarship

Honors and Awards (continued)

2013 – 2015	Howard P. Rawlings Educational Assistance Grant
2013	CCBC Foundation General Scholarship

Service and Outreach

Area Chair

2023	Empirical Methods in Natural Language Processing
------	--

Program Chair

2023	Special Interest Group on Information Retrieval
2023	ICML Workshop Spurious correlations, Invariance, and Stability
2023	Special Interest Group on Information Retrieval
2022 – 2023	International Conference on Learning Representations
2022 – 2023	European Workshop on Reinforcement Learning
2020	Empirical Methods in Natural Language Processing 2020
2020 – 2023	Conference on Neural Information Processing Systems
2020 – 2023	European Chapter of the Association for Computational Linguistics
2020 – 2021, 2023	Annual Conference of the Association for Computational Linguistics
2019 – 2021, 2023	International Conference on Machine Learning
2019 – 2021, 2023	Tapia Conference Scholarship Reviewer
2019	Black in Ai Admission

Workshops

2021 – 2022	Co-chair, Interactive Learning for NLP workshop @ ACL and Neurips
2021 – 2022	Co-chair, NYU AI Winter School for underrepresented groups
2021	Co-chair, Black in Ai at AAAI

Co-curricular

2023	Participated, Cornell NextGen Professor
2022	Volunteered, Ithaca College Central NY LSAMP symposium
2022	Volunteered, WiNLP Panel at NAACL
2022	Volunteered, UMD Doctoral Career Pathways
2020	Participated, CMD-IT Academic Careers Workshop
2020	Volunteered, Bitview (teaching high schoolers computer science)
2020	Volunteered, Industry Mentor UMBC CWIT
2017	Volunteered, Maryland Institute for Minority Achievement and Urban Education College/Career Conference
2015 – 2017	Volunteered, MSDE CTE/PLTW Conference Student Panel
2013 – 2016	Volunteered, Western Tech High Open house

Invited Talks

Brantley, K. *Efficient Policy Optimization Techniques for LLMs* MIT. 2024.

Brantley, K. *Learning from Interaction* George Washington University. 2024.

Brantley, K. *Learning from Interaction* Emory University. 2024.

Brantley, K. *Learning from Interaction* Stony Brook University. 2024.

Brantley, K. *Learning from Interaction* Columbia University. 2024.

Brantley, K. *Learning from Interaction* Rice University. 2024.

Brantley, K. *Learning from Interaction* Harvard University. 2024.

Brantley, K. *Learning from Interaction* Rutgers University. 2024.

Brantley, K. *Learning from Interaction* Princeton University. 2024.

Brantley, K. *Learning from Interaction* Microsoft Research - NYC. 2024.

Brantley, K. *Learning from Interaction* University of Michigan. 2024.

Brantley, K. *Learning from Interaction* Northeastern. 2024.

Brantley, K. *Learning from Interaction* University of Illinois Chicag. 2024.

Brantley, K. *Expert-in-the-Loop for Sequential Decisions and Predictions* Cornell NLP Seminar. 2023.

Brantley, K. *Expert-in-the-Loop for Sequential Decisions and Predictions* Google Brain Montreal. 2023.

Brantley, K. *Ranking with Long Term Constraints* Wayfair. 2023.

Brantley, K. *Reinforcement Learning from Guided Feedback: Addressing the Shortcoming of RL in NLP* Rochester Institute of Technology. 2023.

Brantley, K. *Reinforcement Learning from Guided Feedback: Addressing the Shortcoming of RL in NLP* NC State University. 2023.

Brantley, K. *Expert-in-the-Loop for Sequential Decisions and Predictions* University of Maryland UMIACS. 2021.

Brantley, K. *Successor feature sets: Generalizing successor representations across policies* Microsoft Research Summit. 2021.

Brantley, K. *Active Imitation Learning with Noisy Guidance* The New York Academy of Sciences. 2020.

Brantley, K. *Learning through interaction with experts* Adobe Research. 2020.

Brantley, K. *Non-monotonic sequential text generation* Black in AI Workshop at ACL. 2020.

Teaching Experience

Course Instructor

2020 Bitview (teaching high students computer science)

Teaching Assistant

2019 Computational Linguistics 1

Miscellaneous Experience

Certifications

2013 **Object-Oriented Programming Certificate Community College of Baltimore County** . The certificate is designed for career programmers who wish to add programming language to their skills.

Technical Reports

2017 **Brantley, K.**, Li, J., Guo, F. A Study of Deep Reinforcement Learning for Sentence Compression

2017 Sharaf, A., **Brantly, K.**, Saadatpanah, P., Shafahi, A.. Recurrent Neural Network for Sequence-to-Sequence Reinforcement Learning

2017 **Brantly, K.**, Suarau, B., Mosgin, S. Experimental Comparison of Density-Based Spatial Clustering of Applications with Noise (DBSCAN) using k-d Trees vs. using VP Trees

2015 **Brantly, K.** Suarau, B. Comparing the effect of compression using PCA and DAE on classifier performance in predicting Epileptic Seizures from EEG data

Non-Research Employment

2010 – 2017 **US Department of Defense, Data Scientist.**
Former Positions:
– Cyber Data Analyst:
– DevOps Engineer
– Software Engineer, Distributed Dataflow
– Software Engineer, Testing Engineer
– Software Engineer, Hadoop Engineer
– High School Work-Study, Summer Internship

2010 **Social Security Administration, Summer Intern.**

2010 **Baltimore Convention Center, Waiter**

2010 **McDonald, Cashier**

2007 – 2009 **Green House Cafe, Prep Cook**

References

Available on Request